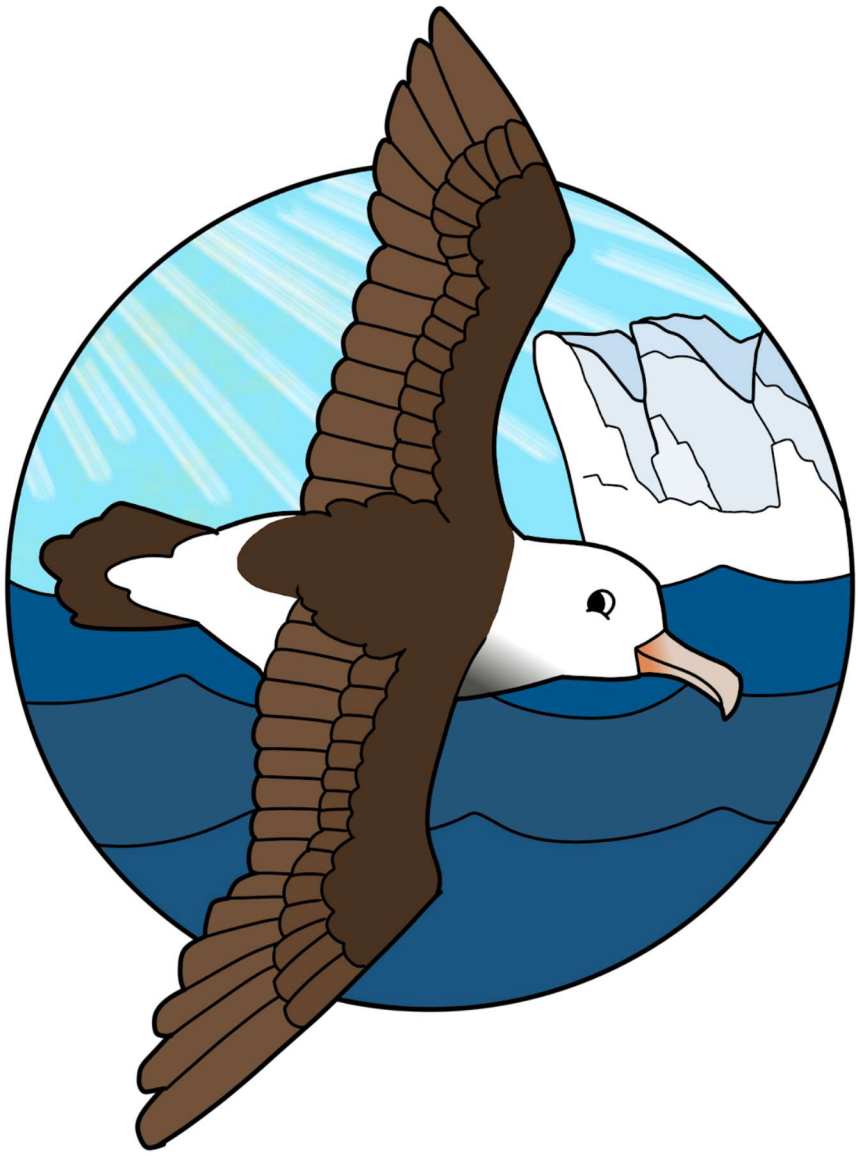


BIRDSO Dynamic Planet Division C TEST PACKET

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CLASS SET DO NOT WRITE

Directions: Answer questions (70); score out of 356. Just get it right. How hard can that be?

This is a pretty difficult test. Best of luck! You'll do great!

Critical Zone MSQ/MCQ: [25]

1. [Multiple Select] Select all of the following choices that properly match the rock/sediment type with a descriptor in terms of porosity and permeability [5 pts]

a. Clays; aquiclude
b. Fractured Granite; aquifer
c. Fresh Granite; aquifuge
d. Shale; aquifuge
e. Fresh Diorite; aquiclude
f. Sandstone; aquiclude
g. Siltstone; aquifuge
h. Tillite; aquifer
i. Gravels; aquitard
j. Limestone; aquifer
k. Sandy clay; aquitard
l. Sandy gravel; aquifer

2. [Multiple Choice] Hydraulic conductivity values can help determine the refraction of flowlines through different media. What equation relates the two properties? [2 pts]

a. $\frac{K_1}{K_2} = \frac{tan(\theta_1)}{tan(\theta_2)}$
b. $\frac{K_1}{K_2} = \frac{sin(\theta_1)}{sin(\theta_2)}$
c. $\frac{K_1}{K_2} = \frac{cos(\theta_1)}{cos(\theta_2)}$
d. $\frac{K_1}{K_2} = \frac{sin(\theta_2)}{sin(\theta_1)}$

3. [Multiple Choice] An aquifer existing above the water table in a certain region would be best described as which of the following? [2 pts]

a. Confined Aquifer
b. Perched Aquifer
c. Unconfined Aquifer
d. Capillary Fringe

4. [Multiple Choice] To what level will an artesian well rise to? How does this level change with distance away from the recharge zone? [2 pts]

a. Piezometric surface; stays constant
b. Water table; decreases
c. Piezometric surface; decreases
d. Water table; stays constant

5. [Multiple Select] The rate of soil moisture loss is dependent on which of the following? [2 pts]
   a. Soil type
   b. Precipitation
   c. Temperature
   d. Humidity

6. [Multiple Select] Which of the following refers to regions that are periodically or permanently unfrozen in permafrost regions? [2 pts]
   a. Yedoma
   b. Talik
   c. Palsa
   d. Active Layer

7. [Multiple Choice] What reef-forming organism effectively filters water in estuaries and decreases the concentrations of pharmaceutical pollutants and increases the clarity of estuarine waters? [2 pts]
   a. Hermatypic Corals
   b. Ahermatypic Corals
   c. Kelp
   d. Oysters

8. [Multiple Choice] Which equation accurately describes hydrologic balance? [2 pts]
   a. P = ET
   b. P < ET + R + I
   c. P = ET + I
   d. P >= ET + R

9. [Multiple Select] Darcy's law is valid for which of the following conditions? [2 pts]
   a. Turbulent Flow
   b. Laminar Flow
   c. Slow flow velocities
   d. Fluid is practically incompressible

10. [Multiple Select] The shape of a soil moisture retention curve is governed by which of the following? [2 pts]
   a. Distribution of pore sizes
   b. Specific Surface
   c. Particle Shape
   d. Distribution of particle sizes

11. [Multiple Choice] The curves for wetting and drying of soils may exhibit some degree of hysteresis. What does this mean? [2 pts]

a. Soil moisture tends to increase before precipitation
b. At a certain pressure head, the amount of soil moisture depends on if the soil is drying or wetting
c. Soil moisture linearly increases during wetting and decreases exponentially
d. Soil moisture increases logarithmically and decreases exponentially

Critical Zone Free Response: [55]

1. [Short Answer] How does infiltration rates change during a period of continuous rainfall? How does it change following a long drought? Elaborate why for each. [7 pts]

2. [Short Answer] What affects the height of the capillary fringe above an aquifer? Why? [5 pts]

3. [Short Answer] How does the installation of a dam change the base level of a river? What effect would this have upstream and downstream of the dam? [6 pts]

4. [Short Answer] What main nutrient pollutants lead to eutrophication? How are these pollutants introduced into fluvial and lacustrine systems? [4 pts]

5. [Short Answer] In what two ways does power generation lead to thermal pollution of rivers and lakes? How may this be remedied? How is this sometimes beneficial? [6 pts]

6. [Short Answer] Given an intruding salt water density of 1.030g/cm^3, freshwater being assumed to be 1.000 g/cm^3, how many **meters below sea level** is saltwater expect to be found in a region where the water table is 5 meters above sea level? [4 pts]

7. [Short Answer] What type of pingo tends to be able to consistently grow larger? What happens near the tops of taller pingos? Explain how both the growth and change near the top happens. [6 pts]

8. [Short Answer] How does patterned ground form? What are the main processes involved in making these features in periglacial environments? [2 pts]

9. [Short Answer] Refer to image C1 and answer the following questions [3 pts each]

a. What does this trend line show between shales and sandstones?
b. What explains the trends of both and the difference between trends for shale and sandstone?

10. [Long Answer] Discuss the effect of drying soils with particular emphasis on clays. Why are certain clays especially detrimental to infrastructure during periods of drought or intense rainfall? How do surface clays change when dried out? What ripple effects may this contribute to? Discuss the wilting point and the effect on a region's flora. [9 pts]

Karst and Lakes Image Questions: [28]

1. Refer to image K1 and answer the following questions. [3 pts each]
   a. Explain how sinkholes may form in volcanically active regions
   b. Explain the shape of the bedrock in Michigan. What happened to this region?
   c. Explain the shape of the limestone deposits in the southern portion of the east coast.

2. Refer to image K2 and answer the following questions. [1, 4, 2 pts]
   a. Name this type of formation.
   b. Explain how this formation was created.
   c. What is the likeliest chemical formula of minerals making up this formation in a cave?

3. Refer to image K3 and answer the following questions. [1 pt each]
   a. What polymorph of calcium carbonate is this feature most likely made of?
   b. What is the name for this formation? (multiple answers acceptable)
   c. What is the exhibited crystal habit of these features?
   d. What non-carbonate (but calcium containing) mineral can form similar features?

4. Refer to image K4 and answer the following questions. Note that in this cross section, A is a layer of cap rock, B is a layer of limestone, C is a layer of basalt. D, F, H are all sandstone layers, while E, G, and I are clayey layers. Layers D, F, H, and B are hydrologically connected aquifers. [2 pts each]
   a. What karstic feature is depicted at the star?
   b. What would happen to a stream that fell through that region depicted by the star?
   c. At what number(s) would you expect springs to exist along the rock?
   d. What best describes unconformity between C and all the tilted layers of sedimentary rock?

Karst and Lakes MCQ/MSQ: [30.5]

1. [Multiple Select] Select all of the following terms correctly matched to their description. [3 pts]
   a. Grike: widened solution fissures found in limestone pavement
   b. Runnel: Concave areas carved out by water channeling and flowing downwards; commonly found on sides of grikes
   c. Mogote: Rounded tower of karstic material in an alluvial plain
   d. Polje: Large, flat floored depression in karstic regions
   e. Doline: Sinkhole
   f. Pseudokarst: Adjective describing features that resemble karstic formations but have different origins

2. [Multiple Select] Which of the following are possible ways of sinkhole formation? [3 pts]
   a. Caprock collapses into a hole in the bedrock
   b. Bedrock collapses to reveal a hole
   c. Cover material collapses into a hole in the bedrock
   d. Caprock sags into bedrock
   e. Cover material washes into a narrow fissure in bedrock
   f. Fissures and corrosion of bedrock creates an excavated region

3. [Multiple Select] Select all of the following lakes that are in exorheic basins. [4 pts]
   a. Great Salt Lake
   b. Lake Michigan
   c. Lake Superior
   d. Lake Baikal
   e. Lake Victoria
   f. Mono Lake
   g. Caspian Sea
   h. Crater Lake

4. [Multiple Select] Select all of the following lakes that are accurately matched to their type. [2.5 pts]
   a. Lake Tanganyika; Tectonic
   b. Lake Titicaca; Volcanic
   c. Lake Vostok; Glacial
   d. Lake Powell; Organic
   e. Lake Tahoe; Anthropogenic

5. [Multiple Select] Which of the following types of lakes would generally be stratified during summer? [3 pts]

a. Warm Monomictic
b. Cold Monomictic
c. Dimictic
d. Meromictic
e. Cold Polymictic
f. Amictic

6. [Multiple Select] Which of the following types of lakes would generally be stratified during winter? [3 pts]

a. Warm Monomictic
b. Cold Monomictic
c. Dimictic
d. Meromictic
e. Warm Polymictic
f. Amictic

7. [Multiple Choice] Which of the following types of lakes would experience overturning in summer? [2 pts]

a. Warm Monomictic
b. Dimictic
c. Cold Monomictic
d. Amictic

8. [Multiple Select] Select all of the following that may trigger a seiche in a lake. [2 pts]

a. Earthquake
b. Nearby, quickly moving storm
c. Strong winds with a large fetch
d. Landslide

9. [Multiple Choice] What would the Volume Development index of a trumpet-shaped lake be? [2 pts]

a. < 1.0
b. > 1.0
c. < 0
d. > 0, <1

10. [Multiple Choice] A lake that is highly resistant to overturning would most likely have what value of relative depth? [2 pts]

a. < 0.0 %
b. < 1.0%
c. ~ 2.0 %
d. > 3.0 %

11. [Multiple Choice] Sphagnum mosses tend to dominate what type of wetland? [2 pts]

a. Swamp
b. Marsh
c. Mudflat
d. Bog

12. [Multiple Select] Select all of the following that accurately match the wetland to the dominant vegetation. [2 pts]

a. Swamp; woody plants
b. Marsh; Herbs and grassy plants
c. Bog; mosses
d. Mudflat; Herbs and grassy plants

Karst and Lakes Free Response: [36]

1. [Short Answer] Why do dry valleys appear dry? What allows these formations to be seemingly fluvial in nature yet devoid of running water? [5 pts]

2. [Short Answer] What causes bogs and fens to generally have quite different pHs? How does this affect their general distributions with respect to the water table? [3 pts]

3. [Short Answer] Why do ice shoves and ice tsunamis happen primarily along the Great Lakes or near the coast and not in smaller bodies of water? [2 pts]

4. [Short Answer] Would a lake take longer to cool down or warm up (same temperature change to some intermediate temperature — 10, 20 degrees C to 15 degrees C)? Is this dependent on the target and initial temperature? Provide a full answer for the cases that are probable in nature. [6 pts]

5. [Short Answer] The index of basin permanence divides the volume of the lake by its shoreline length. What does this information tell us? What would the main differences be between a lake with a small volume and a large shoreline compared to a large volume and a small shoreline? [5 pts]

6. [Short Answer] What importance do wetlands have for the integrity of coastlines? Why are the flora in these regions so irreplaceable (with flora from further inland)? [6 pts]

7. [Long Answer] What ecological benefit and remedial effects do wetlands have for general ecosystem water? Discuss how wetlands help with water quality by remediating various pollutants. [9 pts]

Flowing Water Image Questions: [53]

1. Refer to image F1 and answer the following questions. [2 pts each]
    a. What type of river is the main water body on the map?
    b. What types of wetlands are present on the map?

2. Refer to image F2, representing rivers R and S. [1, 2 pts]
    a. What is this graph representing?
    b. Which river is more prone to flash flooding?

3. Refer to image F3 and answer the following questions. [2 pts each]
    a. What is the name of this diagram?
    b. What diagram(s) would be used to depict this information in a more pictorial manner?
    c. What would lake samples plotted in regions 5 and 8 tell you about the concentration of carbonic acid vs hydrochloric and sulfuric acid?
    d. Which anion(s) are dominant in lake samples plotted in region G?
    e. Which cation(s) are dominant in lake samples plotted in region A?
    f. A volcanic caldera lake with excess sulfuric acid and negligible carbonate and bicarbonate would be found in which region(s)?

4. Answer the following questions regarding hydrograph curves. [4 pts each]
    a. Why does urban development change the hydrograph curve?
    b. Why is this change to the hydrograph detrimental to the environment?
    c. What change(s) can be made to the environment to modify the hydrograph curve to more closely align with the pre-development curve? Give at least one.

5. Answer the following short answer questions concerning meltwater on glaciers. [2 pts each for abc, 3 pts each for d, e]

a. What feature would supraglacial meltwater travel through to become englacial meltwater?
b. What drainage pattern do supraglacial meltwater systems tend to form? Keep in mind the slope of a glacier and the material the meltwater cuts through.
c. About how much surrounding ice can be warmed by one gram of meltwater refreezing? (LH_ice ~ 334 J/g; HC_ice ~ 2090 J/kg)
d. How does ice change as it goes from cold to warm?
e. How does warm ice change the hydrologic conductivity of the glacier?

6. Answer the following short answer questions concerning river evolution. [2 pts each]

a. A straight riffle-pool sequence may develop meanders. What part develops the meanders and why does this make sense?
b. Measuring across a riffle zone may yield a smaller discharge than a pool. Where does the extra water flow from?
c. Where does degradation of the river bed primarily take place?
d. Where does aggradation of the river bed primarily take place?
e. In which direction do knickpoints travel?

Flowing Water MCQ/MSQ: [21.5]

1. [Multiple Choice] Scrolls are best described as old iterations of what features of a meander? [2 pts]

a. Cutbank
b. Oxbow Lake
c. Point Bar
d. Meander Neck

2. [Multiple Select] Select all of the following that would promote meandering and prevent braiding of a river. [2 pts]

a. Vegetated floodplain
b. Consolidated sediment
c. Loose alluvium as river bed
d. Seasonal flow of water

3. [Multiple Choice] Hack's law that relates the length of the longest river to the area of a catchment basin is best described as what type of relationship? [2 pts]

a. Power law
b. Linear relationship
c. Exponential relationship
d. Step function

4. [Multiple Choice] In Strahler ordering of a normal, mature river system, what is the expected ratio between the number of streams of each successive Strahler order (aka. Bifurcation ratio) ? (1st order: 2nd order, 2nd:3rd, etc.) [2 pts]

a. 0 - 1
b. 1 - 2
c. 2 - 4
d. 5 - 10

5. [Multiple Select] Select all of the following that accurately describe various measured properties of drainage systems. [2.5 pts]

a. Stream frequency has units of 1/area
b. The elongation ratio, circularity ratio, and bifurcation ratio are dimensionless
c. A high drainage density value may indicate rather low infiltration rates
d. Mature rivers tend to have higher stream length ratios
e. Drainage texture has units of 1/length

6. [Multiple Choice] How many years would pass before a 100 year flood had a 50% probability to occur? [2 pts]

a. ln(0.50)/ln(0.99)
b. ln(1/0.5) * 100
c. ln((100/50)^100)
d. ln(0.50)^50

7. [Multiple Select] Select all of the following terraces correctly matched to their nature of origin. [2 pts]

a. Strath Terrace: Erosional, bedrock
b. Fill Terrace: Depositional, alluvium
c. Cut Terrace: Erosional, alluvium
d. Nested Fill terrace: Depositional, alluvium

8. [Multiple Select] Select all of the following that accurately describe the prevalent ions in freshwater (averaged out; using mg/L). [2.5 pts]
   a. Carbonate generally dominates instead of bicarbonate
   b. Nitrate generally dominates instead of sulfate
   c. Potassium generally dominates instead of Sodium
   d. Boron is found at higher concentrations than silica
   e. Chloride is found at higher concentrations than fluoride

9. [Ordering] Order the following from best sorted to least sorted. [4.5 pts]
   a. Alluvial Fan
   b. Loess
   c. Delta
   d. Glacial till

Flowing Water Free Response: [107]

1. Answer the following questions concerning the hypothetical drainage system in image F4. [2 pts each]

Using Strahler Ordering:
   a. How many first order streams are there?
   b. What is the order of the trunk stream?
   c. How many river segments are third order streams?

Using Horton Ordering:
   d. What is the order of the trunk stream?
   e. How many third order streams are there?
   f. How many fourth order streams are there?

Using Shreve Ordering:
   g. How many first order streams are there?
   h. What is the stream order of the trunk stream?
   i. How many third order streams are there?

Not concerning ordering:
   j. What type of drainage pattern does this most closely resemble?
   k. What is the slope of this region? What is the nature of the bedrock?

2. Given the measurements in F5, solve for the slope of this channel, assuming the top is open to the air (not wetted) [5 pts]

3. [Short Answer] What is the discharge of the channel described in F5 when filled and tilted to a slope of 0.08? [4 pts]

4. [Short Answer] What is the recurrence interval of a flood that has a 33.63% chance of occurring at least once within the next 33 years? [4 pts]

5. [Short Answer] A boat floats in a river flowing at 2m/s; the boat moves **half as fast** as the water. In 20 minutes, the boat travels a straight line distance of 1 km. What is the sinuosity of the river? [5 pts]

6. [Short Answer] The measured length of a river is 527.8 kilometers and the averaged sinuosity is 1.55. When split into two sections of 263.9 km each, the averaged sinuosity of both is 1.33. What is the angle between the straight line distance from the midpoint to either endpoint of this river? [5 pts]

7. [Short Answer] Going from the headwaters to the river mouth, how would you expect each of the following to change? [2 pts each]
    a. River Gradient
    b. Hydraulic Radius
    c. Manning's Coefficient
    d. Discharge
    e. Average Flow Velocity
    f. Ultimate Base Level

8. [Sketch] Draw the flow lines of groundwater near both rivers flowing in image F6. Numbers indicate hydraulic head[4 pts]
    a. Which stream would run for longer during the dry season, assuming one or both of the streams dries up? [2 pts]

9. [Short Answer] A series of channels were cut into a river to increase its flow velocity by straightening its course. The river used to have a sinuosity of 1.62 but now has a sinuosity of 1.32. Its current length is around 250 km. If each channel on average decreased the river's length by 0.80 km, how many meanders were bypassed? [3 pts]

10. [Short Answer] (A bit of an involved question) Alice is given the task to design a semicircular canal with a maximum flow rate of at least 100 m^3/s to be a useful emergency spillway for a small dam. Slopes steeper than 0.08 will damage any sort of concrete she uses. Relief is 20 meters. The walls of her canal should always be 0.50 meters thick when measured perpendicular to the slope it is on. [22 pts total]

    Refer to F7 for a diagram.

    She may choose to use three types of concrete with varying smoothness:

Polished concrete: n = 0.010 for $150/m^3
Cement finished concrete: n = 0.012 for $130/m^3
Wood-formed concrete: n = 0.015 for $125.00/m^3. Cost should be minimalized.

a. What size (give inner radius) and with what concrete should Alice design for? What is the final cost? (Use three significant figures (r = x.xx; cost = $xxx,000)) [16 pts]

Alice decides to build some tolerance into the system and plans for a channel with a 2 meter inner radius since the Manning constant for the concrete is not always even across the length of the entire constructed channel. During testing, they find that at capacity, the flow rate is approximately 150 m^3/s.

b. What was the Manning Constant of their channel? [4 pts]

c. Alice believes the concrete was poured such that the Manning constant should be close to what was measured, if not slightly smoother (due to water erosion). What type of concrete did they use for the channel? [2 pts]

11. [Short Answer] A hemispherical lake with a radius of 1.00 km sits in a small depression at the top of a hill. There is no water inflow, but there is a small river fed by the lake with an average discharge of 0.50 m^3/sec. The lake is bounded near the bottom by an aquitard such that groundwater flow away from the lake is essentially horizontal. The watershed is a circle centered on the center of the lake with a radius of 3.00 km. Refer to image F8 for a sketch. Answer the following questions concerning this lake. [19 pts total]

a. What is the yearly average precipitation into the watershed to maintain equilibrium with the discharge of the lake via the small river, assuming no evapotranspiration, and negligible exchange with groundwater? [6 pts]

b. How much should the yearly average precipitation be if the lake remains in equilibrium while the runoff is decreased by 40% due to transpiration and infiltration, and 1.0% of the lake's volume is evaporated each year? [4 pts]

c. What will the pH of the lake be, with respect to other nearby effluent lakes? [1 pts]

d. The lateral component of area of a hemisphere is given by the equation $\frac{(\pi * r)^2}{2}$
If the average hydraulic gradient, i, is 0.02, the yearly precipitation is 2.5 meters, and assuming the lake is in equilibrium with the previous assumptions given in b, what is the permeability of the confining sediment? [8 pts]